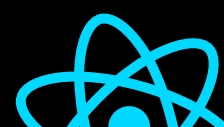


React Context API

Complete Guide



Context API

React Context API, introduced in React v16.3, provides a powerful mechanism for sharing data across component trees without prop drilling.

This eliminates the need to manually pass props down through every level of the component hierarchy, making your code cleaner and easier to maintain.



When to Use Context API

Context API is ideal for sharing data that is:

- Global: Authentication state, user preferences, theme settings.
- Used by Deeply Nested Components: Data required by components separated by many levels in the component tree.
- Read-Only: Data that components only need to consume, not modify.



1

1. Creating the Context:

We use

`React.createContext(defaultValue)` to create a context object.

The `defaultValue` is the initial value the context will hold.



2

2. Providing the Context:

Wrap the part of your component tree that needs access to the context with a `Context.Provider` component.

Pass the data you want to share as a prop to the `Provider`.



3

3. Consuming the Context:

In components that need to access the context data, use the `useContext` hook.

`useContext(ContextName)` will return the current context value.




```

import React, { createContext, useState } from 'react';

const ThemeContext = createContext('light');

const ThemeProvider = ({ children }) => {
  const [theme, setTheme] = useState('light');

  const toggleTheme = () => {
    setTheme(theme === 'light' ? 'dark' : 'light');
  };

  const value = { theme, toggleTheme };

  return (
    <ThemeContext.Provider value={value}>{children}</ThemeProvider>
  );
};

export { ThemeContext, ThemeProvider };

// App.js

import React from 'react';
import { ThemeContext, ThemeProvider } from './ThemeContext';

const Header = () => {
  const { theme } = useContext(ThemeContext);
  return (
    <header style={{ backgroundColor: theme === 'light' ? 'white' : 'black' }}>
      <h1>My App</h1>
      <button onClick={() => {}}>Some Button</button>
    </header>
  );
};

const App = () => {
  return (
    <ThemeProvider>
      <Header />
    </ThemeProvider>
  );
};

export default App;

```





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